

The Full-Dimensional Linear Cahn–Hilliard Spinodal Atlas

Route-A source-local certificate HCS-C304

3 September 2026

Abstract

For every finite $d \geq 1$, $\kappa > 0$, and $\alpha \in \mathbb{R}$, we classify the linear Cahn–Hilliard flow $u_t = -\kappa \Delta^2 u - \alpha \Delta u$ on the 2π -periodic torus. The result gives the self-adjoint analytic trace-class semigroup, exact lattice-shell spectrum, and the complete stable/critical/spinodal atlas. It also closes the energy law, Morse index and kernel, all fastest-shell ties, actual-support asymptotics, recurrence, and the singular $\kappa = 0$ face.

1 One theorem for the entire parameter space

Let $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, $H = L_0^2(\mathbb{T}^d)$, and

$$A = -\kappa \Delta^2 - \alpha \Delta, \quad D(A) = H^4(\mathbb{T}^d) \cap H, \quad \kappa > 0, \quad \alpha \in \mathbb{R}.$$

Write $r_d(n) = \#\{k \in \mathbb{Z}^d : |k|^2 = n\}$.

Theorem 1 (Full linear spinodal atlas). *For every finite $d \geq 1$ the operator A is self-adjoint, bounded above, and has compact resolvent. It generates a self-adjoint analytic semigroup $S(t)$, trace class for every $t > 0$. On shell $n > 0$,*

$$Ae_k = \sigma_n e_k, \quad \sigma_n = \alpha n - \kappa n^2, \quad (1)$$

with multiplicity $r_d(n)$. The mean-zero equilibrium is exponentially stable iff $\alpha < \kappa$, critical iff $\alpha = \kappa$, and spinodally unstable iff $\alpha > \kappa$. Its unstable Morse index and kernel dimension are

$$M = \sum_{n < \alpha/\kappa} r_d(n), \quad K = \sum_{n = \alpha/\kappa} r_d(n), \quad (2)$$

where absent shells contribute zero. At criticality $K = r_d(1) = 2d$.

The spectral bound is attained precisely on

$$\mathcal{N}_* = \operatorname{argmax}_{n \geq 1, r_d(n) > 0} (\alpha n - \kappa n^2), \quad (3)$$

including every tie. For $u_0 \neq 0$, let $\lambda_ = \max\{\sigma_n : P_n u_0 \neq 0\}$. Then*

$$e^{-\lambda_* t} S(t) u_0 \longrightarrow \sum_{\sigma_n = \lambda_*} P_n u_0 \quad \text{in } L^2. \quad (4)$$

Every recurrent state is stationary; hence there is no nonstationary periodic solution.

Proof. For $e_k = (2\pi)^{-d/2} e^{ik \cdot x}$, $\Delta e_k = -|k|^2 e_k$, so (1) follows. The real diagonal values tend to $-\infty$ as $|k| \rightarrow \infty$, proving self-adjointness, compact resolvent, and an upper bound. The spectral theorem yields the analytic semigroup, while

$$\sum_{k \neq 0} e^{t(\alpha|k|^2 - \kappa|k|^4)} < \infty \quad (t > 0)$$

by quartic domination, proving trace class. The sign factorization $\sigma_n = n(\alpha - \kappa n)$ proves the chamber and index formulas.

The maximization in (3) is over the whole represented spectrum, not a numerical cutoff. Indeed

$$\sigma_n = \frac{\alpha^2}{4\kappa} - \kappa \left(n - \frac{\alpha}{2\kappa} \right)^2. \quad (5)$$

If $\alpha/\kappa \leq 1$, then for $n > 1$, $\sigma_n - \sigma_1 = (n-1)[\alpha - \kappa(n+1)] < 0$. If $\alpha/\kappa > 1$, shell one has positive rate whereas every $n \geq \alpha/\kappa$ has nonpositive rate. Thus an explicit finite set of represented integers contains every maximizer, and all ties are retained. For a fixed datum the same decay at infinity gives a largest rate in its actual support. Factoring $e^{\lambda_* t}$ and dominated convergence prove (4). Finally, recurrence along $t_j \rightarrow \infty$ forces $e^{t_j \sigma_n} \hat{u}_0(k) \rightarrow \hat{u}_0(k)$ coefficientwise; every nonzero coefficient must have $\sigma_n = 0$, so the state is stationary. $\square$

2 Gradient-flow law and singular faces

Set

$$\mathcal{F}(u) = \frac{1}{2} \int_{\mathbb{T}^d} (\kappa |\nabla u|^2 - \alpha |u|^2) dx, \quad \mu = -\kappa \Delta u - \alpha u.$$

The equation is $u_t = \Delta \mu$. Periodic integration by parts gives the exact dissipation identity

$$\frac{d}{dt} \mathcal{F}(u(t)) = \langle \mu, \Delta \mu \rangle = -\|\nabla \mu\|_2^2. \quad (6)$$

This explains the energy signature without turning linear instability into a nonlinear saturation or coarsening theorem.

Proposition 1 (Boundary completion). *On full L^2 , the constant mode is stationary and equals the conserved mean. On the mean-zero space H , the singular face $\kappa = 0$ is interpreted with its natural generator domain: for $\alpha \neq 0$, the operator $-\alpha \Delta$ has domain $H^2(\mathbb{T}^d) \cap H$, while for $\alpha = 0$ the zero generator has domain all of H . It gives forward heat when $\alpha < 0$, the identity when $\alpha = 0$, and no bounded L^2 C_0 semigroup when $\alpha > 0$.*

Proof. The zero shell has rate zero. On the stated natural domains, the $\kappa = 0$ positive shells have rates αn . Negative α gives heat, zero gives identity, and positive α makes $\sup_n e^{t\alpha n} = \infty$ for every $t > 0$. $\square$

Face	Exact conclusion
$\alpha < \kappa$	every mean-zero shell decays exponentially.
$\alpha = \kappa$	only shell one is neutral; its dimension is $2d$.
$\alpha > \kappa$	finitely many represented shells are unstable.
fastest tie	retain the entire tied spectral projection.
missing ambient leader	use the largest rate in the actual support of u_0 .
$\kappa \downarrow 0$	heat / identity / no bounded semigroup according to the sign of α .

Source lineage

Cahn and Hilliard introduced the nonuniform free-energy model, and Cahn’s spinodal paper is the historical owner token for the instability language. Elliott and Zheng provide classical analytical context. These references establish lineage only.

References

- [1] J. W. Cahn and J. E. Hilliard, “Free Energy of a Nonuniform System. I. Interfacial Free Energy,” *J. Chem. Phys.* 28 (1958), 258–267. DOI: [10.1063/1.1744102](https://doi.org/10.1063/1.1744102).
- [2] J. W. Cahn, “On spinodal decomposition,” *Acta Metallurgica* 9 (1961), 795–801. DOI: [10.1016/0001-6160\(61\)90182-1](https://doi.org/10.1016/0001-6160(61)90182-1).
- [3] C. M. Elliott and S. Zheng, “On the Cahn–Hilliard equation,” *Arch. Rational Mech. Anal.* 96 (1986), 339–357. DOI: [10.1007/BF00251803](https://doi.org/10.1007/BF00251803).